

A100-SXM4-40GB | Qwen3-30B-A3B MoE layer | free persistent kernel (no build cost charged) | 45 MHz grid, every pair enumerated.

Baseline: the kernel as deployed -- clocks unlocked, PyTorch launches it, the GPU picks its own frequency. Measured 6x: it settles at 1322 MHz, 3.759 ms, 1504 mJ.

16 cold experts ($\varphi=0.045$)

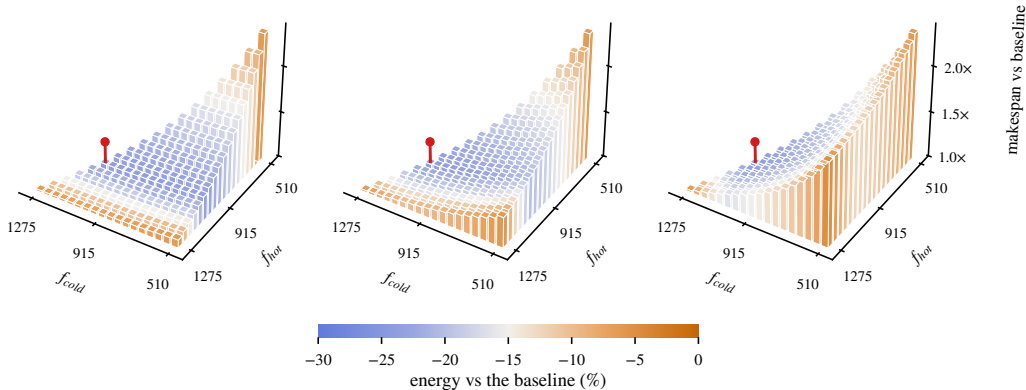
48 cold experts ($\varphi=0.208$)

96 cold experts ($\varphi=0.596$)

best -23.9% energy at 1.22× latency

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Height is what the design costs, colour what it buys. Every pillar stands above 1.0: nothing anywhere is faster than the baseline.

The best pillar reaches -24% energy at 1.22× latency -- but ONE fixed clock at 1035 MHz, no split and no new kernel, is -24.1% at 1.23×.

The split is not what saves the energy; lowering the clock is. The red pin marks each panel's best pillar.